

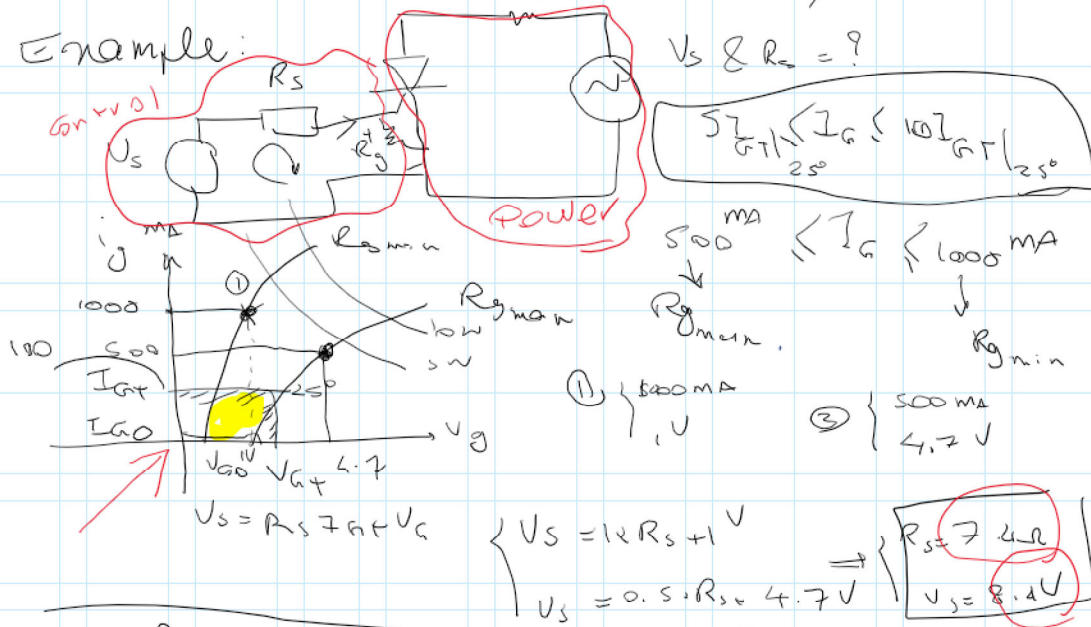
Recall: Thyristor/SCR $\begin{matrix} A \\ \nearrow \\ K, T \end{matrix}$

① V_{DEN}/V_{REM}

② $I_T(A)$

③ phase controlled / inverter grade / LAT

Example:

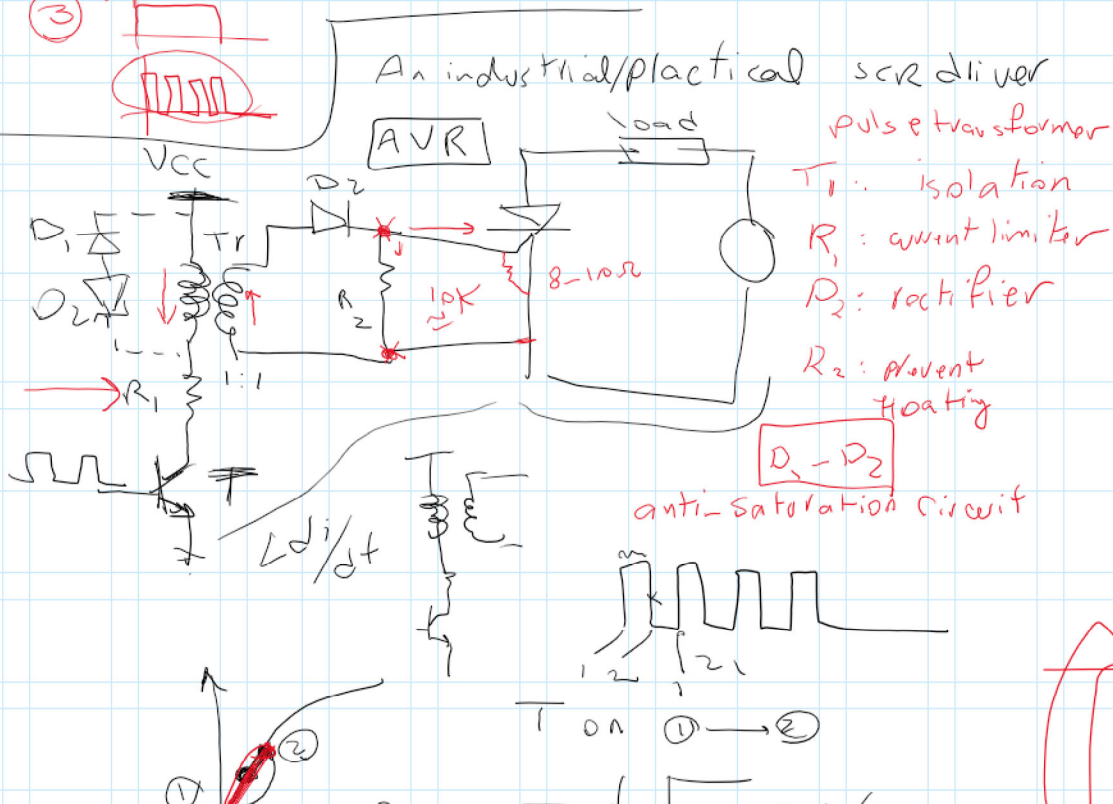


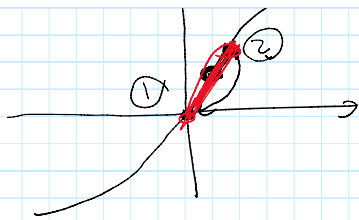
Specifications of a good SCR driver

① isolation

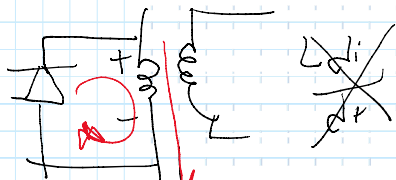
② controllable

③

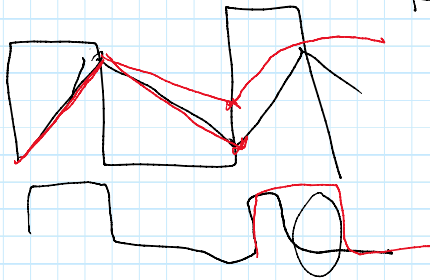




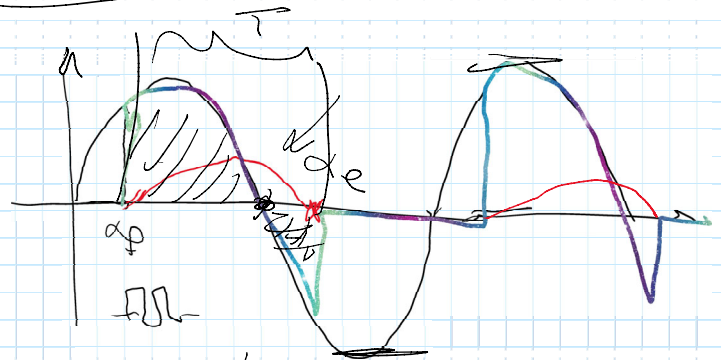
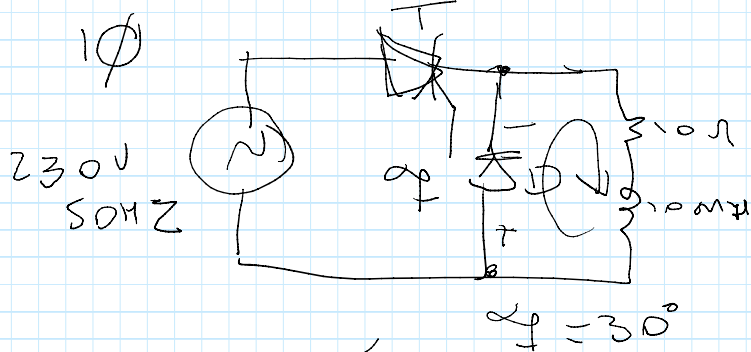
I on ① → ②



1 → 2 $V = 10V$
 2 → 1 $V = -2V$
 $V = -10V$



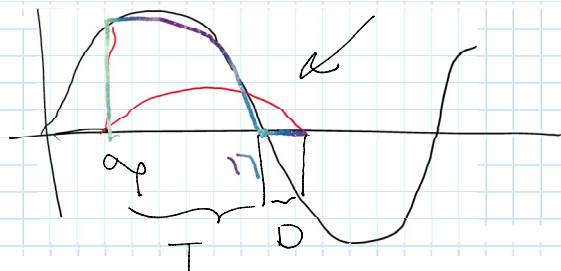
Thyristor phase-controlled circuits.



$$\bar{V}_0 = \frac{1}{2\pi} \int_{\alpha_f}^{\alpha_e} V_m \sin \omega t d\omega t = \frac{V_m}{2\pi} (\cos \alpha_f - \cos \alpha_e) \quad \text{without diode}$$

$\alpha_f = \phi \rightarrow$ Diode

with the diode

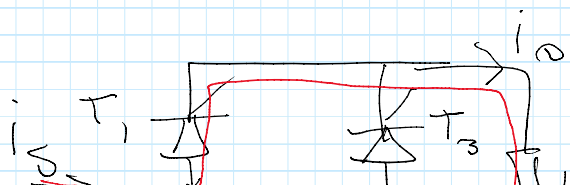


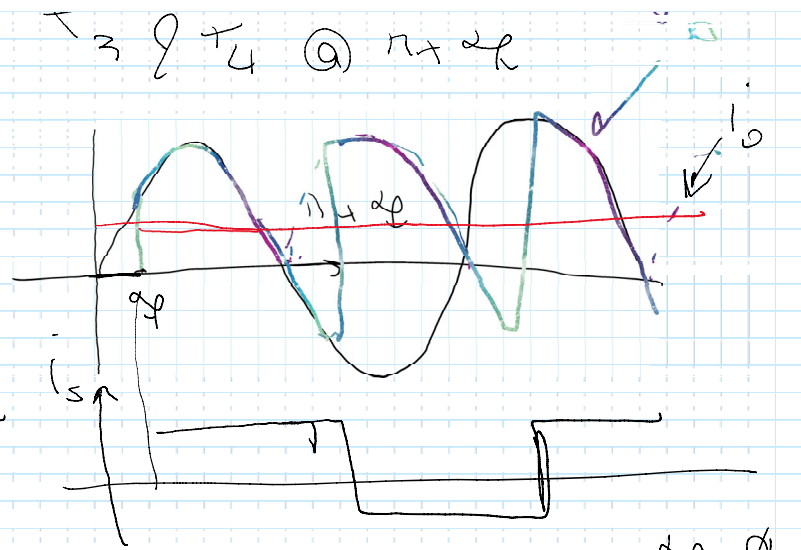
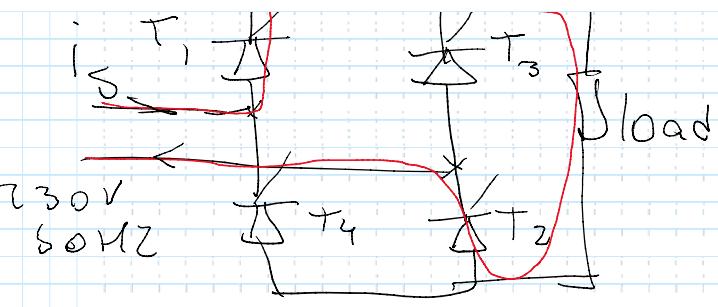
Phase-controlled / $V_{PRM} \text{ or } V_{RRM} \geq \text{s.p. } 230\sqrt{2}$

$T_T \text{ (AV)} \geq \text{s.p. } I_T \quad \left\{ \begin{array}{l} \alpha_f = \phi \end{array} \right.$

$T_1 \text{ or } T_2 \text{ @ } \alpha_f$

$T_3 \text{ or } T_4 \text{ @ } \pi + \alpha_f$



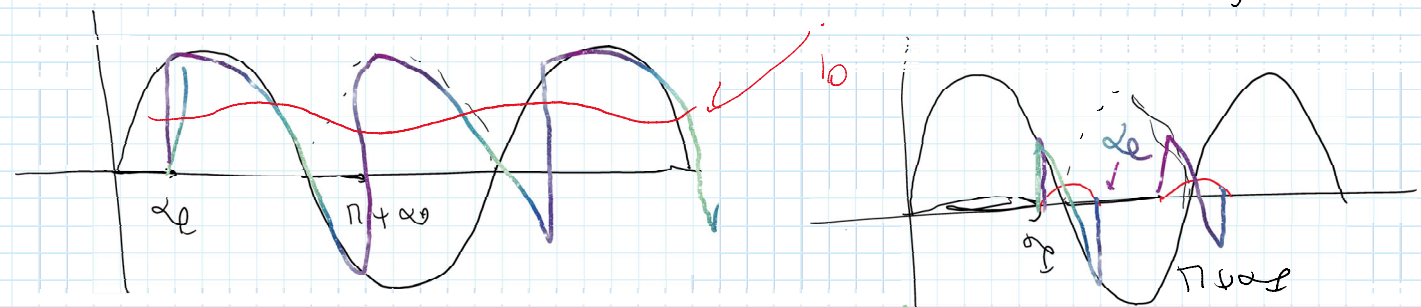


assumption: continuous load current

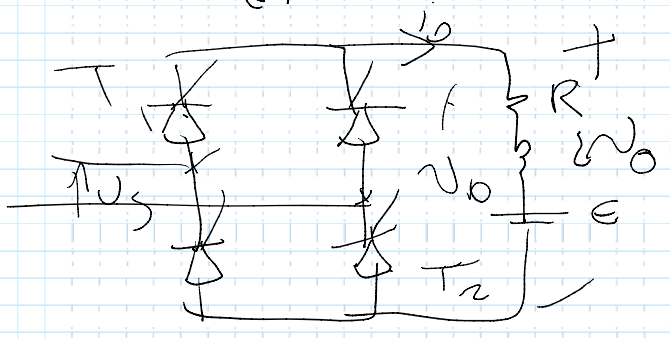
$$\bar{V}_0 = \frac{1}{\pi} \int_{\alpha_f}^{\pi + \alpha_f} V_m \sin \omega t d\omega t = \frac{2V_m}{\pi} \cos \alpha_f$$

assumption: RL load

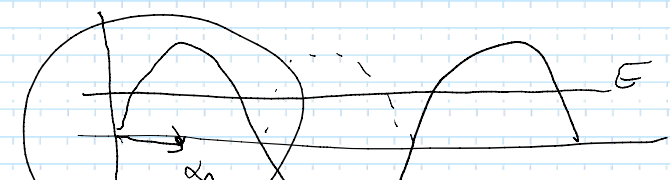
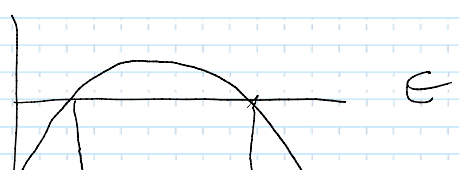
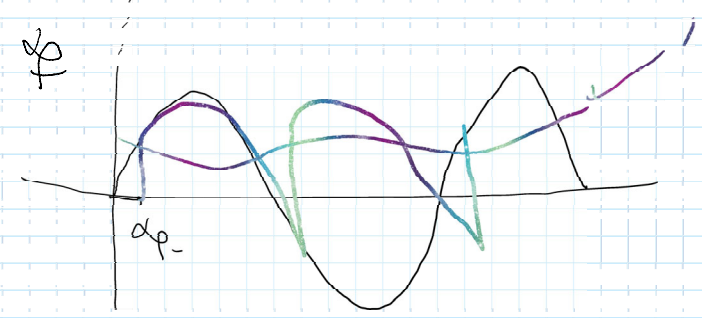
- continuous (small α_f)
- discontinuous (large α_f)

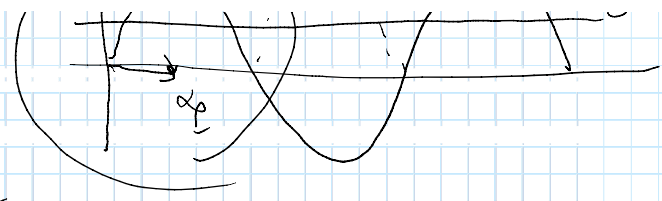
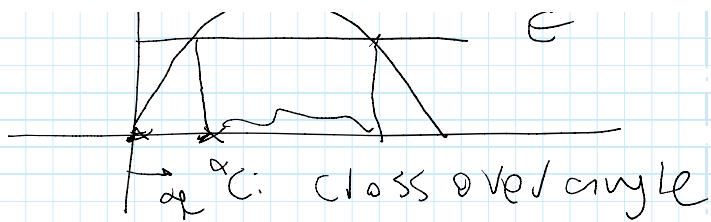


$$Ri_o + L \frac{di_o}{dt} = V_m \sin \omega t$$



$$V_o = \frac{1}{\pi} \int_{\alpha_f}^{\pi + \alpha_f} V_m \sin \omega t d\omega t$$





feasible firing angles $\alpha_e > \alpha_c$ $\alpha_c < \alpha_e < \pi - \alpha_c$

$$\bar{V}_0 = \frac{1}{\pi} \left\{ \int_{\alpha_p}^{\alpha_e} V_m \sin \omega t d\omega t + \int_{\pi+\alpha_e}^{\pi+\alpha_p} E_c d\omega t \right\}$$

(α_e)

6-pulse thyristor / based phase controlled 3φ rectifier

